

Mechanisms underlying T cell ageing

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Abstract | T cell ageing has a pivotal role in rendering older individuals vulnerable to infections and cancer and in impairing the response to vaccination. Easy accessibility to peripheral human T cells as well as an expanding array of tools to examine T cell biology have provided opportunities to examine major ageing pathways and their consequences for T cell function. Here, we review emerging concepts of how the body attempts to maintain a functional T cell compartment with advancing age, focusing on three fundamental domains of the ageing process, namely self-renewal, control of cellular quiescence and cellular senescence. Understanding these critical elements in successful T cell ageing will allow the design of interventions to prevent or reverse ageing-related T cell failure.

Quiescence

A phase in the cell cycle in which the cell is not dividing or preparing to divide but still has the ability to do so in the presence of an appropriate signal. Quiescent cells have low metabolic activity and reduced protein synthesis.

Senescence

A cellular state in which an irreversible growth arrest programme has been initiated that limits the lifespan of the cell and prevents unlimited cell proliferation. It can be caused by replication-induced telomere shortening or DNA damage. In contrast to quiescent cells, senescent cells have a secretory phenotype and upregulated protein synthesis.

The adaptive immune system has been a prime focus of research on the ageing process and its implications^{1,2}. Ageing-associated changes to the immune system are clinically important, leaving older individuals more vulnerable to new infections and to reactivation of latent viruses. Aggravating this problem is the fact that many of the current vaccine strategies only induce incomplete protection in older populations³. Improving vaccine responses is paramount for healthy ageing. This goal is achievable, as recently exemplified by the development of an adjuvanted varicella zoster virus (VZV) vaccine that is effective irrespective of age⁴. However, further progress will require approaches tailored to the ageing immune system and, therefore, a better knowledge of the specific immune defects. Strategies in young individuals cannot be simply translated to the older population, as shown by a recent meta-analysis of influenza virus vaccination studies⁵, in which biomarkers that were predictive of a superior vaccine response in the young were no longer informative in older individuals and an inflammatory signature had a positive effect in young individuals but was harmful in older adults⁵.

In addition to the implications for immune system function, studies on T cell ageing provide a unique opportunity to explore the fundamental mechanisms that drive the ageing process in general⁶. The T cell system has unique mechanisms of replenishment, with the production of new T cells entirely dependent on thymic activity, which rapidly declines during adolescence and early adulthood⁷. In the absence of thymic output, naive T cells essentially function as their own stem cells. The T cell system is also an excellent model to study the influence of ageing on cell population dynamics⁸. Immune competence is determined by the frequencies of T cells that recognize one particular antigenic peptide. Therefore, the population has to establish a balance between maintaining a highly diverse set of

T cell specificities in sufficient frequencies to be able to respond and increasing the clonal size of the T cell specificities that are needed to control acute, chronic and latent infections over the lifetime of the individual. Finally, T cells are a model system enabling studies of cellular states relevant for ageing, including cellular quiescence, senescence and exhaustion^{9–12}.

Here, we review T cell ageing with respect to these mechanistic phases of the ageing process, focusing mainly on data available from human studies. By analogy to the stem cell theory, which postulates that the ageing process results from the inability of stem cells to replenish a tissue with functionally competent cells, we discuss whether and how the T cell population is maintained with age. Moreover, we discuss whether T cell ageing reflects cellular senescence or the failure of cells to maintain quiescence and instead undergo differentiation. We highlight how the T cell ageing process is influenced by the accumulation of DNA damage and programmed pathways, in particular those that drive cell differentiation or senescence.

T cell replenishment in immune ageing

Naive T cell generation by peripheral T cell self-renewal. One hallmark of ageing is the decline in homeostatic and regenerative capacity that is common to all tissues and organs and generally related to stem cell ageing^{6,13–15}. T cell replenishment in adult humans is special in that it is, at least in part, uncoupled from stem cells, relying less on thymic activity and more on homeostatic self-renewal of naive T cells. The generation of nascent T cells is entirely dependent on the thymus, where progenitor cells differentiate and are positively and negatively selected to generate the repertoire of self-restricted, self-tolerant and functional T cells. However, unlike any other organ, the thymus undergoes involution during childhood and adolescence, leading

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Exhaustion

Refers to an impaired ability of effector T cells to carry out their functions, such as cytotoxicity and cytokine secretion, owing to chronic stimulation by antigen.

Homeostatic proliferation

A process of activation and proliferation of lymphocytes in the lymphopenic environment. T cell homeostatic proliferation is driven by T cell receptor interactions with self-peptide–MHC and T cell responses to cytokines such as IL-7, IL-15 and possibly IL-21.

Virtual memory T cells

Antigen-inexperienced memory-phenotype T cells, which may be induced by T cell receptor cross reactivity, low-affinity peptide and/or MHC ligands and certain cytokines.

Fibroblastic reticular cells

(FRCs). Specialized reticular fibroblasts located in the T cell areas of lymph nodes and other secondary lymphoid organs. They provide IL-7 for T cell survival and produce collagen-rich reticular fibres and form stromal networks and conduits that are important for the trafficking of immune cells.

to reduced numbers of thymocytes and thymic epithelial cells and disruption of the tissue architecture^{7,16}. Thymic export rates decrease rapidly between the ages of 1 and 8 years and decrease at a more gradual rate in subsequent years¹⁷.

The consequences of reduced thymic activity for maintaining a naive T cell compartment are remarkably species specific. In young mice, thymic output is at least 2–4 times higher than the daily production of T cells by peripheral T cell proliferation¹⁸. As a consequence, naive T cell numbers cannot be maintained with declining thymic activity. Moreover, molecular defects in aged haematopoietic stem cells (HSCs) in mice mirror those in aged peripheral T cells, and T cell function in mice can be improved by reversing HSC defects such as by inhibiting CDC42 activity in aged HSCs¹⁹. By contrast, even in young adult humans, the majority of T cells derives from peripheral T cell proliferation and not from the thymus (FIG. 1). The contribution of the thymus to T cell generation is estimated to decline from only ~16% to <1% over the adult lifetime^{20,21}, a decline that can be compensated by minimal adjustments in homeostatic proliferation. This is consistent with the observation that turnover of peripheral naive T cells does not noticeably increase with age, possibly with the exception of the very elderly^{22,23}. Moreover, in contrast to mice, so far there is no evidence that defects in HSCs translate to T cell defects in humans. For example, mutations of the epigenetic modifiers *TET2* and *DNMT3A*, which have been associated with HSC clonality, are found frequently in the elderly²⁴ and could be detrimental to T cell diversity, have not been reported in peripheral T cells, possibly because low thymic activity prevents the generation of such T cells from mutant stem cells.

CD8⁺ T cells: the Achilles' heel of T cell homeostasis.

Homeostatic proliferation in humans is sufficient to maintain a sizable naive CD4⁺ T cell compartment (FIG. 1). By contrast, loss of circulating naive CD8⁺ T cells with age is more severe in terms of both relative and absolute cell numbers^{25–27}. In fact, reduced numbers of naive circulating CD8⁺ T cells is the most consistent and

prominent marker of immune ageing in healthy older adults, independent of comorbidities and chronic latent infections²⁸. The diminution of naive CD8⁺ T cells may even be underestimated, because CD8⁺ central memory T cells lose CD45RO expression and regain CD45RA expression after resolved viral infections, thereby masquerading as naive T cells²⁹ (FIG. 2). Consistent with this finding, a subset of naive CD8⁺ T cells is able to produce effector cytokines upon stimulation, indicating that this subset is primed³⁰. It is not clear why there is reduced resilience of the naive CD8⁺ T cell compartment to age; it can be assumed that naive CD8⁺ T cells receive more consistent survival signals than naive CD4⁺ T cells based on the facts that MHC class I molecules are ubiquitously expressed and CD8⁺ T cells are responsive to IL-15 in addition to IL-7. It is possible that IL-2 supports better homeostatic survival of naive CD4⁺ T cells, as CD4⁺ T cells but not CD8⁺ T cells gain expression of CD25 (the α -subunit of the high affinity IL-2 receptor) with age³¹. Because they are more easily activated, naive CD8⁺ T cells may also be depleted through apoptosis or assume a memory phenotype. Such cytokine-activated CD8⁺ T cells — known as virtual memory T cells — accumulate in mice with age and have been shown to develop features of cellular senescence^{32–34}. Mouse naive CD8⁺ T cells are more susceptible to this homeostatic proliferation-induced differentiation into virtual memory T cells than naive CD4⁺ T cells. It is therefore possible that the generation of virtual memory T cells accounts for the preferential erosion of the naive CD8⁺ T cell compartment. A cell population in humans that has similarities to virtual memory T cells in mice has been described³⁵; however, it is not known whether virtual memory T cells also accumulate in humans with age.

The tissue niche for homeostatic proliferation.

Peripheral homeostasis of T cells depends on their recruitment to secondary lymphoid organs, where they encounter IL-7 produced by fibroblastic reticular cells (FRCs) and low-level T cell receptor (TCR) stimulation through antigen-presenting cells displaying self-antigens^{36,37}. Impaired access to or changes in the architecture of secondary lymphoid organs with age could therefore have a negative impact on the maintenance of the naive T cell compartment (FIG. 2). Interestingly, this has indeed been found in aged mice, independent of the production of IL-7 by FRCs^{38,39}. The importance of FRC networks for T cell homeostasis has also been demonstrated in humans, although not yet in the context of ageing. Initial studies showed that the inflammation associated with HIV infection leads to lymph node fibrosis and contributes to peripheral T cell depletion⁴⁰. In subsequent studies, this phenomenon was also seen in human populations that have a high incidence of chronic or recurrent infections other than HIV⁴¹. Lymph node fibrosis inversely correlated with peripheral naive T cell counts as well as with immune responses to yellow fever vaccination, indicating the functional relevance of this process⁴¹.

In addition to failing FRC networks, diminished IL-7R expression on T cells contributes to failure in T cell

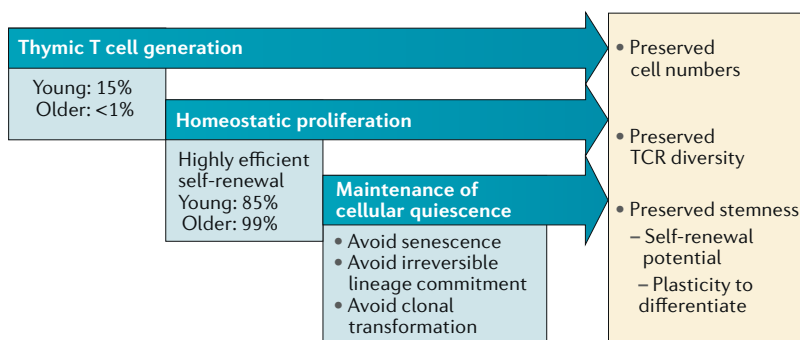


Fig. 1 | Successful T cell replenishment by self-renewal. Throughout adult life, most T cell replenishment occurs by self-renewal of peripheral T cells rather than de novo generation by thymic activity. In healthy ageing, homeostatic proliferation is efficient to maintain a large compartment of naive T cells and is sufficiently stochastic to preserve a highly diverse T cell receptor (TCR) repertoire. Signals driving homeostatic proliferation should be below the threshold that interferes with cellular quiescence and initiates differentiation.

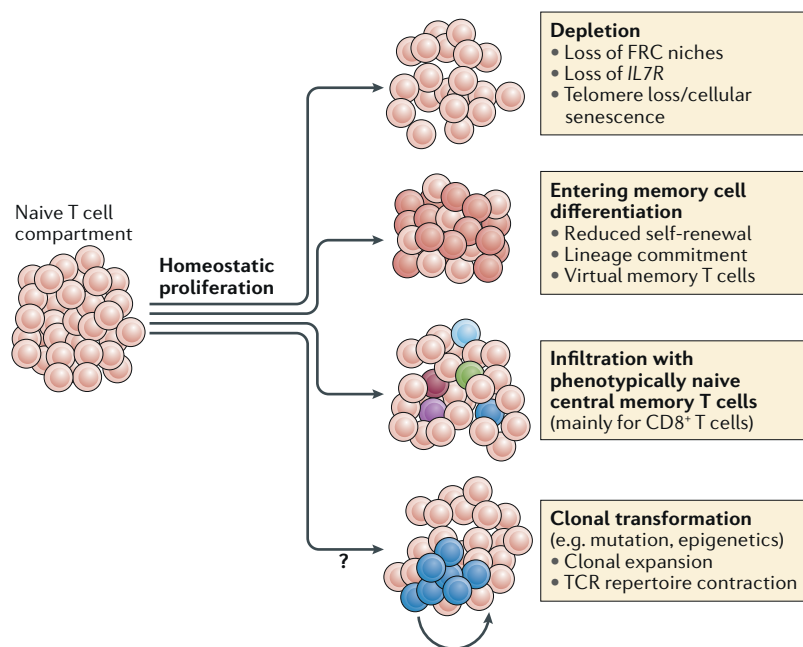


Fig. 2 | Mechanisms compromising T cell homeostasis. Maintenance of a large, diverse and functional naive T cell compartment can be compromised by several mechanisms. Homeostatic proliferation can be impaired through T cell-intrinsic defects or failing niches formed by fibroblastic reticular cell (FRC) networks in the T cell zones of lymph nodes, leading to a reduction in compartment sizes. For unknown reasons, CD8⁺ naive T cells are more prone to homeostasis failure. If quiescence is not maintained, naive T cells undergo differentiation towards memory T cells. Whether differentiation proceeds as far in humans as seen with virtual memory cells in mice is unclear, but markers of early differentiation states are frequently observed, even in healthy older adults. Again, aged CD8⁺ T cells appear more susceptible to differentiation. Moreover, the naive T cell compartment can be infiltrated by memory T cells that masquerade as naive T cells and compete for niches. Phenotypic switches regularly occur in CD8⁺ central memory T cells, but are uncommon for CD4⁺ T cells. Finally, clonal expansion of selected T cells may occur because of changes in growth behaviour. Such transformations are common for haematopoietic stem cells, but they have not been widely studied for T cells. Computer modelling predicts that they could result in rapid contraction of repertoire diversity. TCR, T cell receptor.

homeostasis with age. In genome-wide studies of ageing-associated epigenetic changes, reduced chromatin accessibility at the *IL7R* locus in CD8⁺ T cells was identified as an ageing hallmark⁴². In addition, microRNA networks in CD8⁺ T cells from older individuals are altered, reducing the activity of forkhead box protein O1 (FOXO1), the key transcription factor regulating *IL7R* gene expression⁴³. Interestingly, compared with CD8⁺ T cells, CD4⁺ T cells show less epigenetic alteration with age, which may explain their higher resilience⁴².

Maintaining T cell diversity: how much is enough?

T cell homeostasis is unique in that it needs to maintain not only population size but also high diversity of TCR specificities (FIG. 1). The notion that T cell specificities are lost with age, resulting in a contraction of the TCR repertoire, has been an attractive explanation for defective immune responses. However, there is little evidence that 'holes' in the repertoire occur in humans without extreme reduction in population sizes. In computer simulations of human T cell homeostasis, only minimal contraction in diversity was observed over

50 years of homeostatic proliferation despite the absence of thymic production; a marked reduction in diversity was not even observed when the size of the naive T cell compartment shrunk by >50%, as is regularly the case for CD8⁺ T cells⁴⁴. Clonal extinction is more likely if the initial clonal size is very small, which does not appear to be the case in humans. Moreover, peripheral T cell selection implies that homeostatic proliferation occurs in TCR-specific niches, where clones compete for the recognition of self-antigen in the context of self-MHC; however, experimental evidence for the existence of such niches is lacking²¹.

Given the enormous diversity of the human naive TCR repertoire, clonal size estimates are difficult. Theoretical estimates have been in the order of ten cells per clone⁴⁵, which is likely at the lower end for humans, in particular for those clones that are generated early in life and have space to expand. Frequencies of naive T cells containing TCR excision circles in newborns suggest clonal sizes of 100, which would make complete extinction of a clone over a lifetime an unlikely event⁴⁶. Several studies have used next-generation sequencing to estimate repertoire contraction with age. Due to undersampling and contamination with memory cells, estimates of naive T cell diversity are frequently misjudged^{47–50}. In part circumventing these limitations by analysing replicate samples of purified T cell subsets, we estimated that up to 10⁸ unique TCR β-chain sequences exist in young adults⁵¹. TCR β-chain diversity declined with age, but only by a factor of 3–5 in healthy elderly individuals, which still leaves a very diverse repertoire. Whether this decline is of functional significance is unclear.

Clonally expanded T cells with a fitness advantage challenge diversity.

In silico simulation of T cell homeostasis provided evidence that, under certain conditions, selection of peripheral T cell clones on the basis of 'fitness' compromises repertoire diversity⁴⁴. Rapid contraction in TCR diversity was observed if single clones changed their growth behaviour and gained the ability to rapidly expand (FIG. 2). Acquired mutations conferring clonal growth advantage have been identified in HSCs with age, with *TET2* and *DNMT3A* being frequent drivers of clonal haematopoiesis^{24,52}. As discussed above, thymic T cell generation at older ages is very low, preventing the differentiation of such HSCs into T cells. However, similar to HSCs, aged T cells have an extensive replicative history due to homeostatic proliferation and antigen-induced clonal expansion. It has been recently estimated that there are three mutations for every cell division⁵³. It is therefore likely that T cells, unrelated to HSCs, accumulate mutations that could lead to the generation of T cell specificities with a growth advantage and rapid contraction in repertoire diversity⁴⁴. Indeed, somatic mutations were discovered in clonally expanded CD8⁺ T cells in patients with rheumatoid arthritis⁵⁴.

Oligoclonality is most prevalent in the subset of T effector memory CD45RA cells (T_{EMRA} cells), which are populations of T cells associated with memory inflation following certain viral infections⁵⁵. Clonally expanded

TCR excision circles

Small, stable circles of DNA excised during T cell receptor gene rearrangement in the thymus.

T effector memory CD45RA cells

(T_{EMRA} cells). Terminally differentiated antigen-specific memory T cells that re-express CD45RA. These cells have been identified in both CD4⁺ and CD8⁺ T cell compartments, have short telomeres, exhibit cell cycle arrest, express DNA damage foci and have a secretome reminiscent of senescent cells.

T_{EMRA} cells are enriched in the bone marrow⁵⁶ and therefore do not seem to compete with naive T cells for lymph node niches. In studies comparing memory T cells from the blood, spleen, bone marrow, lymph nodes and lungs, those in the lymph nodes were found to be less differentiated than those in other compartments⁵⁷. It is therefore unlikely that T_{EMRA} cells outcompete normal T cell self-renewal in the lymph nodes.

Memory T cell homeostasis: a surprisingly dynamic process to maintain long-term memory. Regulation of memory T cell populations is highly complex and more information on the dynamics of the memory T cell compartment is required, particularly in humans, to fully understand how it is affected by age. In the most simplified model, memory T cells that are maintained in the absence of antigens have to be distinguished from those that re-encounter antigens. Antigen re-encounter may be periodic, for example, in the setting of reinfection or reactivation of latent viruses, or chronic, for example, by antigens expressed during viral latency or chronic active viral infection.

Studies in settings wherein antigen re-exposure was excluded concluded that immune memory lasts for 40–60 years in the absence of antigen and therefore may be lifelong⁵⁸. Frequencies of antigen-specific memory T cells after small pox vaccination decreased by 50% every 8–15 years⁵⁹. However, the lifetime of individual memory T cells is much shorter; in humans, it is generally ~6 months as determined by deuterium labelling studies⁶⁰. Thus, maintenance of immunological memory is a dynamic process that is not only determined by the survival of individual memory T cells but also by homeostatic proliferation of these cells. Recent studies of antigen-specific CD8⁺ T cell kinetics after yellow fever vaccination have shown that, by 30 days, the cells have entered a quiescent state, which is characterized by a long intermitotic phase and division rate of once every 485 days²⁹. The division rate of this antigen-specific CD8⁺ T cell population was lower than previous estimates for unfractionated CD4⁺ or CD8⁺ memory T cells (both ~180 days); however, it should be noted that the yellow fever-specific CD8⁺ T cells no longer express the classical memory cell markers but are contained within the compartment of cells expressing naive T cell phenotypic markers, thus resembling the subset of cells recently coined as stem cell-like memory T cells^{61–63}. Recent studies of total stem cell-like memory CD4⁺ and CD8⁺ T cell populations yielded self-renewal estimates of ~430 days⁶⁴, which is similar to estimates for yellow fever-specific CD8⁺ T cell self-renewal⁶⁴.

In summary, T cell memory appears to be maintained less by longevity of individual T cells than by homeostatic proliferation, which occurs at a slow rate, but is still faster than self-renewal of naive T cells. Based on data from mice, this proliferation is cytokine driven and has less requirement for tonic TCR signalling^{65–67}. Limited kinetic data from in vivo deuterium labelling of healthy young and older individuals suggest similar turnover rates of central memory and effector memory populations, suggesting that this process is intact in

older individuals, again with the caveat that only healthy individuals were studied^{23,68}. Moreover, TCR repertoire studies of memory cells only showed a small contraction in diversity⁵¹.

However, analysis of turnover of total T cell populations does not take into account possible heterogeneity in population dynamics. Indeed, analytical approaches to deuterium labelling data performed better using multi-exponential rather than single-exponential models, indicating a kinetic heterogeneity that is not reflected in conventional phenotypic markers⁶⁹ and that is typical for memory T cells but not for naive T cells. One possible and intuitively appealing explanation is temporal kinetic heterogeneity — that is, a temporally limited increase in the proliferative rate of a subpopulation due to antigenic re-exposure in the absence of clinical symptoms. Such a proliferative burst could lead to the extinction of unrelated memory T cells and therefore to the loss of T cell memory with age. Surprisingly, the computational analysis of available data is more consistent with the alternative model of kinetic heterogeneity — that is, the existence of memory cell subpopulations with stable distinct homeostatic proliferation kinetics⁶⁰. Such population heterogeneity may, in part, challenge the concept that immune memory is maintained by the division of short-lived memory cells rather than the persistence of long-lived cells. Kinetic heterogeneity has also been described among stem cell-like memory T cells. Although most stem cell-like memory T cells are frequently dividing, a small subpopulation of these cells showed self-renewal dynamics that are consistent with the decline in memory⁶⁴.

Recent studies of uninfected mice kept in a clean environment indicated an even higher complexity of the kinetic structure of the CD4⁺ memory T cell compartment⁷⁰. These studies revealed two central memory and two effector memory CD4⁺ T cell subpopulations of equal sizes, but with vastly different kinetics — one dividing and dying every 3 days, the other dividing every 170 days (of note, the average turnover rates of memory T cells in mice are tenfold higher than in humans). These studies also documented a constant influx of naive T cells into the memory cell compartment, with a surprisingly high replacement of up to 10% of cells per week in young mice in the absence of any obvious antigenic challenge. The replacement rate declined with age, which may reflect declining rates of memory T cell differentiation due to reduced numbers of naive T cells or higher resistance to replacement of memory T cells. Computational modelling could not clearly distinguish between these two models, but favoured the resistance model. These high replacement rates, as well as the existence of a highly proliferative memory T cell subset, represent major challenges to preserving memory.

It will be important to determine whether humans also have a pool of memory T cells that is rapidly and constantly replaced, whether and how this pool contributes to immune memory and, perhaps most importantly for immune health in older individuals, the nature of those memory T cells that appear to resist replacement.

Memory inflation

The gradual accumulation of peptide-specific CD8⁺ T cells with an effector memory phenotype that occurs after the resolution of certain acute viral infections during viral latency (for example, cytomegalovirus infection). Induction in the setting of chronic antigen persistence suggests the clonal expansion is antigen driven and not mutation driven.

Stem cell-like memory T cells

A subset of memory T cells that has naive-like features and phenotypes including enhanced self-renewal and multifunctional capacity.

Longevity of T cell memory can differ depending on viral infection or modes of vaccination and is currently not well understood. The most marked example in the context of ageing is the difference in immune memory to latent herpesviruses. Frequencies of VZV-specific T cells decline with age, resulting in frequent VZV reactivation presenting as shingles^{71,72}, whereas the T cell memory to cytomegalovirus, and in part to Epstein–Barr virus, tends to inflate, contributing to the age-associated increase of effector memory T cells and T_{EMRA} cells^{26,73}. Further insights into the dynamics of the memory T cell population are needed to better understand the determinants of memory longevity.

T cell homeostasis: vulnerable to failure but generally robust. In summary, CD4⁺ T cell homeostatic mechanisms can be surprisingly resilient to withstand the challenges of ageing in healthy individuals, including thymic involution, peripheral selection, changing environments and recurring antigenic stimulation. The notion that T cell memory is maintained by unbiased homeostatic proliferation of memory T cells appears to be an oversimplification as the dynamics in the memory compartment are more complex than previously thought. The influence of age on memory cell dynamics and the implications for interventions to prolong immune memory have not been determined.

Most of the more mechanistic studies done so far have focused on healthy individuals, concluding that the ageing process does not necessarily induce major distortions. How far these findings can be generalized to less healthy individuals remains unclear. While there is an association between frailty and a state of chronic activation of the innate immune system, known as inflammageing, it is unclear whether a similar relationship exists for adaptive immune defects. Poorer vaccine responses and higher risk of latent virus reactivation in frail individuals have been reported by some but not all studies^{74–77}. Mechanistic studies of T cell homeostasis in populations including frail elderly individuals are needed. Larger population studies may also capture more infrequent and stochastic events such as mutations that drive clonal expansion or cause massive contraction in diversity. Moreover, disease states are likely to have a major influence over T cell homeostasis, as shown by the effects of chronic or recurring inflammation on FRC niches.

Failed quiescence in ageing

T cells are quiescent but poised. Naive and memory T cells exist in a quiescent state, poised to proliferate and differentiate upon antigen stimulation. Maintaining quiescence is vital to retain self-renewal potential and differentiation plasticity throughout life (FIG. 1). In quiescence, cell division and growth are coordinately downregulated, cells are arrested in their cell cycle and they have low metabolic and mammalian target of rapamycin complex (mTORC) activity, resulting in reduced ribosome biogenesis and protein synthesis⁷⁸. Disruption of quiescence has been implicated in stem cell ageing¹⁵. For example, activation of transcription factor networks, including basic leucine zipper transcriptional factor ATF-like

(BATF), induce clonality in lymphoid precursors and a preference for myeloid lineage differentiation with age⁷⁹. As observed for FRC networks in the aged lymph node, stem cell niches in young mice generally promote stem cell proliferation more efficiently than those from older mice. However, the aged niche can also become more stimulatory, driving cells out of quiescence. For example, elevated fibroblast growth factor 2 signalling from the aged niche has been associated with depletion of the resident stem cell population in the muscle and diminished muscle regenerative capacity⁸⁰; this model may also apply to HSCs, which have been found to be more activated in the aged niche^{81,82}. Similarly, the favourable quiescent state of T cells is continuously challenged by environmental stimuli, including those that drive homeostatic proliferation, in particular under conditions of lymphopenia.

Differentiation of aged T cells due to failed quiescence

Human T cell ageing shares many features with T cell differentiation, presumably due to a loss of quiescence (FIG. 3). The clearest evidence for this is the accumulation of virtual memory CD8⁺ T cells in aged mice (FIG. 2). These cells develop in response to cytokine stimulation in the absence of a cognate antigen encounter and they express the same marker profile as cells undergoing lymphopenia-induced homeostatic proliferation⁸³. The generation of virtual memory T cells has been considered to be beneficial as these cells show innate properties and provide bystander protective immunity in response to secreted cytokines^{84,85}. However, being more differentiated memory T cells, they have lost proliferative potential and therefore may account for the decline in primary CD8⁺ T cell responses with age³⁴. To what extent human virtual memory T cells accumulate with age is not firmly established. If virtual memory T cells constitute a major proportion of the memory T cell compartment, TCR sequence diversity in the memory T cell population should increase with age, which is not the case. In fact, although virtual memory T cells are mostly CD8⁺ T cells, TCR diversity is fivefold to tenfold less for CD8⁺ memory T cells than for CD4⁺ memory T cells⁵¹.

Epigenetic evidence for loss of stemness in aged T cells

Epigenetic studies have provided evidence that the quiescent state is not fully maintained in human T cells with ageing^{86,87}. Age-related changes in T cell chromatin accessibility are very similar to those that are seen with T cell activation and differentiation (FIG. 3); these changes are much more prominent in CD8⁺ T cells than in CD4⁺ T cells. Sites of increased accessibility in naive CD8⁺ T cells include motifs for basic leucine zipper domain-containing transcription factors, such as BATF and activator protein 1 (AP-1), but not motifs of the T-box or RUNX transcription factors, which is consistent with incomplete differentiation. Progression towards a more differentiated state with age is also observed for CD8⁺ memory T cells and, to some extent, for CD4⁺ naive T cells. At the single cell level, older T cells have increased population heterogeneity in histone acetylation, indicating diversity in activation and differentiation

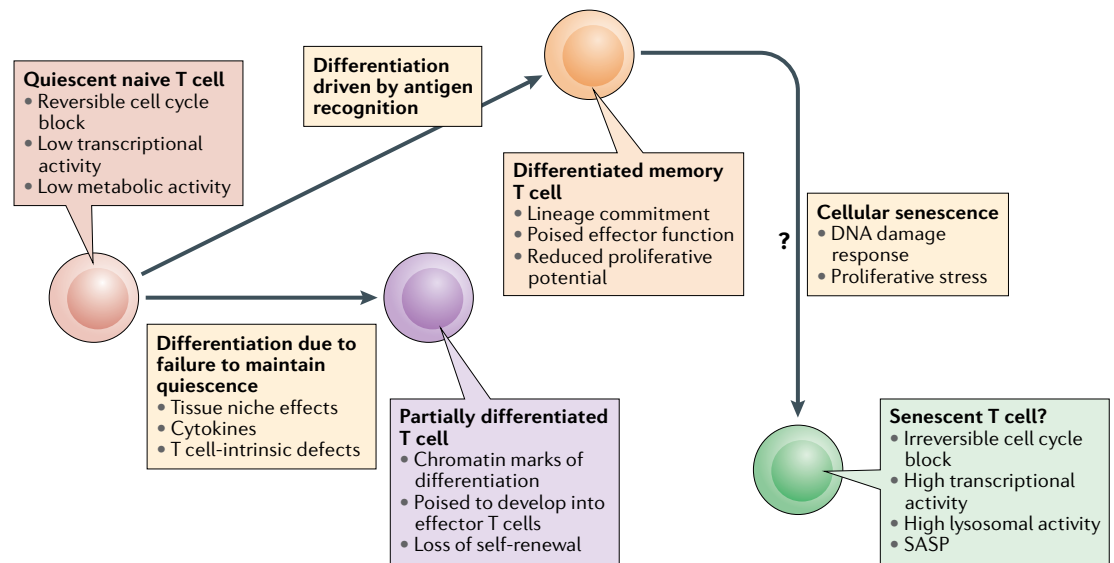


Fig. 3 | Activation of differentiation pathways in T cell ageing. Many findings in T cell ageing are related to differentiation. Cumulative antigenic experience leads to the accumulation of differentiated cells. Even in the absence of antigen stimulation, T cells can leave their usual quiescent state and accumulate as partially differentiated cells. These cells, as well as functionally differentiated memory T cells, regain quiescence. Cellular senescence is fundamentally different from quiescence. SASP, senescence-associated secretory phenotype.

states⁸⁸. Differentiation of naive T cells and/or phenotypic conversion of memory T cells could explain these findings.

Functional evidence for differentiation of aged naive T cells.

The naive T cell compartment is also increasingly recognized as being more heterogeneous than previously thought⁸⁹. Gene expression studies of naive T cells suggest that T cell ageing, at least in part, involves the same pathways observed in T cell activation and differentiation^{31,90}. Single-cell RNA sequencing studies have shown that ageing increases the cell–cell variability in gene expression during early activation of mouse CD4⁺ T cells, possibly due to increased heterogeneity in differentiation states⁹¹. Age-associated changes in microRNA networks in part resemble those of differentiation^{43,92,93}. Flow cytometry studies have shown that the expression levels of some key microRNAs, such as miR-146a, which increases with ageing and differentiation, and miR-181a, which decreases with ageing and differentiation, follow a Gaussian distribution, supporting the idea that the entire population is affected and the observed changes are not due to subset contamination⁴³. Transition to a more differentiated state is sufficient to change cell behaviour without causing typical senescence (FIG. 3). Most interesting is a bias in the differentiation potential of older T cells⁹⁴; for example, increased expression of miR-21 in naive CD4⁺ T cells from older individuals targets the degradation of negative regulators of several signalling pathways, including phosphatase and tensin homologue, Sprouty 1 and programmed cell death 4. The ensuing sustained activation of these signalling pathways favours the generation of inflammatory effector T cells over that of T follicular helper cells and memory precursor cells⁹⁴. Additionally, transcription factor networks in aged memory CD4⁺

T cells are poised to favour effector states and expression of the ectoATPase CD39, which has been associated with defective vaccine responses and effector states, including T cell exhaustion^{95,96}.

Entry into a more differentiated state may also imply a bias in lineage commitment and therefore a loss in plasticity to respond. For example, a decrease in variation of CD38 expression with age among naive CD4⁺ T cells was observed, and it was proposed that the increased heterogeneity in young T cells endows more plasticity in response patterns⁹⁷. Similarly, we observed increased responsiveness to transforming growth factor- β in the presence of transcription factors such as BATF and interferon regulatory factor 4, which favoured the generation of T helper 9 cells in older individuals⁹⁸.

Cellular senescence in T cell ageing

Mechanism of cellular senescence. Cellular senescence is generally implicated as a major mechanism of ageing-associated dysfunction^{6,99}. It is a form of irreversible growth arrest induced in response to telomere shortening or a variety of stress responses, mostly involving DNA damage^{100,101}. To avoid chromosomal instability and to safeguard genomic stability, DNA damage responses are triggered, leading to permanent cell cycle arrest¹⁰². T cells are exposed to short-term and long-term stressors that could induce senescence¹⁰³. Telomere shortening in naive T cells is generally thought to be a consequence of homeostatic proliferation¹⁰⁴. Consistent with their increased replicative history, memory T cells have shorter telomeres than naive T cells¹⁰⁵. Moreover, the DNA damage response is activated in T cells proliferating in response to antigen recognition^{106,107} and is predictive of less successful vaccine responses in older individuals¹⁰⁸.

DNA damage responses

A cell response triggered by DNA damage such as single or double strand breaks. The DNA damage response stops cell cycle progression to enable repair before the damage is transmitted to progeny cells. Checkpoints in the mammalian DNA damage response are controlled by the phosphoinositide 3-kinase-related kinases ATM and ATR.

Lack of typical cellular senescence features in aged naive and memory T cells.

Is cellular senescence functionally relevant for immune ageing in humans? Expression of the DNA damage response component p16^{Ink4a} (also known as CDKN2A) in human peripheral T cells has been described as a biomarker of ageing¹⁰⁹. However, p16^{Ink4a} is also expressed during normal differentiation into effector T cells and is therefore not in itself sufficient to indicate senescence. Chromatin accessibility mapping has shown increased accessibility at the *INK4A-ARF* locus in memory and effector T cells compared with CD8⁺ naive T cells, but this is only slightly more so in older individuals⁹⁰. Additionally, ageing-associated changes in heterochromatin accessibility (similar to changes occurring in senescent cells^{110–112}) are subtle and occurred in CD8⁺ T cells but not in CD4⁺ T cells⁴². Moreover, most aged naive and memory T cells do not display the phenotypic hallmarks of senescent cells⁷⁸ such as a secretory phenotype, increased size, increased lysosomal content and function, and vacuolated and granular morphology. Most importantly, the vast majority of aged naive and central memory T cells is able to proliferate when appropriately activated. In fact, aged naive CD4⁺ T cells from patients with rheumatoid arthritis divide even faster, despite showing accelerated ageing (based on increased telomere attrition and DNA damage levels), compared with T cells from age-matched healthy controls^{107,113–115}.

T_{EMRA} cells: effector T cells or senescent T cells? Although most aged T cells do not exhibit functional defects due to cellular senescence, rare senescent T cells may still contribute to immune defects. In non-immune aged tissues, senescent cells only represent a small fraction of cells. Whether and how such infrequent cells have an impact on tissue function has been the basis for debate. It is now generally accepted that senescent cells, triggered by DNA damage, secrete a range of mediators, such as pro-inflammatory cytokines, that have potent paracrine and endocrine effects, a property known as senescence-associated secretory phenotype (SASP)^{116–120}. This function of senescent cells can be beneficial, as long as it is of limited duration and senescent cells are rapidly cleared by the immune system. Long-term persistence of SASP can result in chronic inflammation and disease.

T_{EMRA} cells meet several of the criteria for cellular senescence¹²¹ (FIG. 4), including short telomeres, exhibiting cell cycle arrest, expressing DNA damage foci and having a secretome reminiscent of SASP. In common with exhausted T cells, T_{EMRA} cells develop in the setting of chronic viral infection and exhibit cell cycle arrest; however, unlike exhausted T cells, they maintain high effector functionality and actually increase their production of inflammatory mediators¹². How these T cells accumulate in chronic viral infection is not entirely clear, but it may be due to failure in the attrition of effector

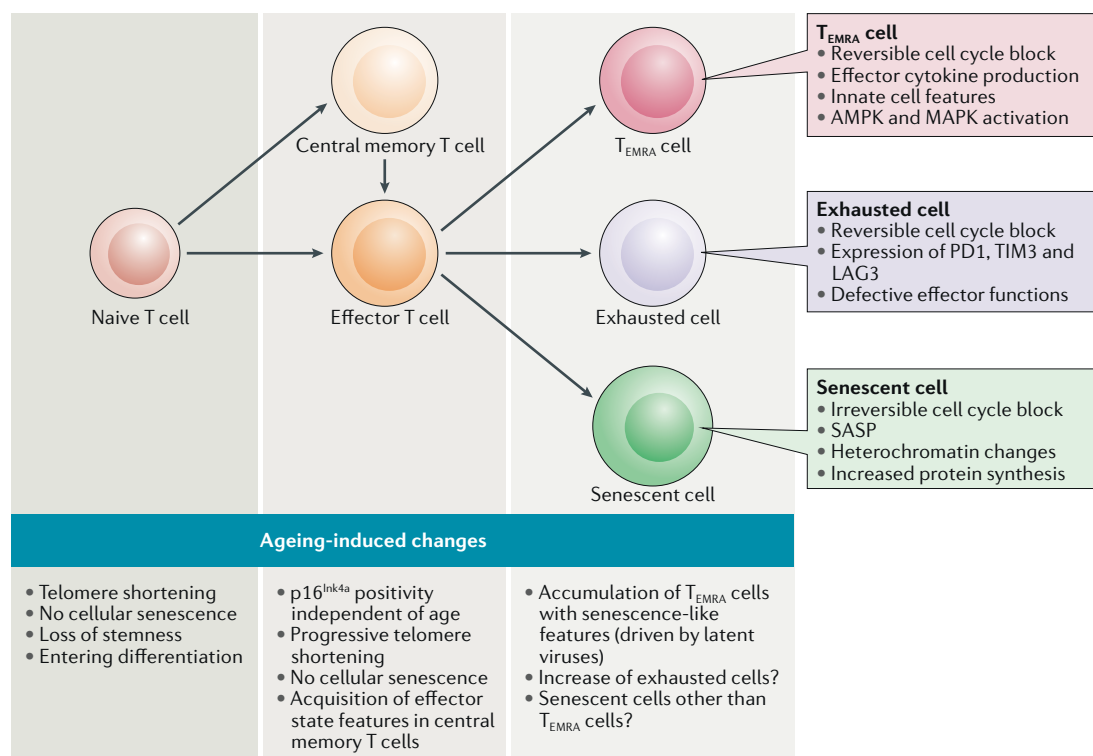


Fig. 4 | Relationship between T cell differentiation and cellular senescence in T cell ageing. Following activation and differentiation, naive and memory T cells activate pathways that are also involved in developing cellular senescence such as telomeric erosion, activation of DNA damage responses and expression of cell cycle inhibitors. However, true cellular senescence is not observed in most naive and memory T cells. T effector memory CD45RA cells (T_{EMRA} cells) and exhausted T cells exhibit cell cycle blocks that are still, at least in part, reversible, distinguishing them from truly senescent cells. The excessive cytokine production by T_{EMRA} cells is reminiscent of the senescence-associated secretory phenotype (SASP), whereas exhausted T cells lack effector functions, setting them apart from T_{EMRA} cells and senescent cells. AMPK, AMP-activated protein kinase; MAPK, mitogen-activated protein kinase.

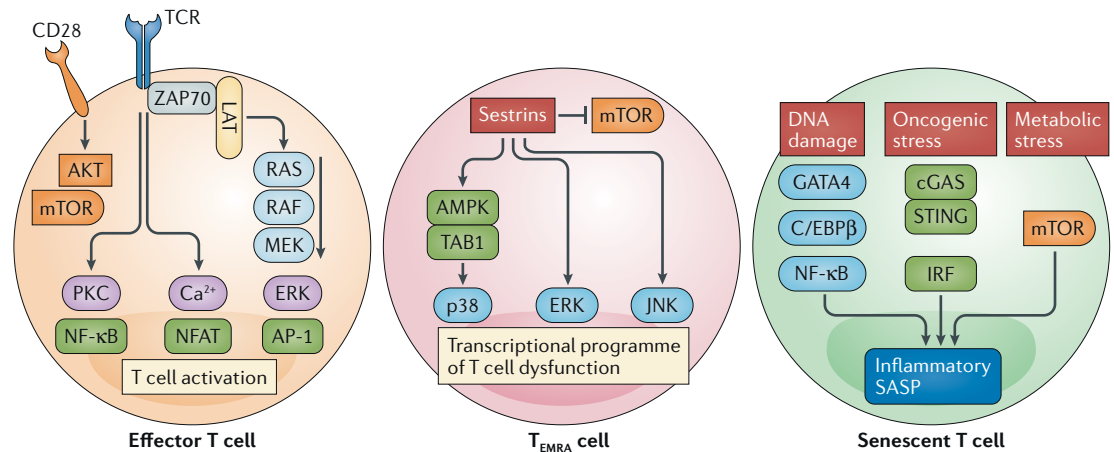


Fig. 5 | Regulation of cytokine transcription in effector T cells, T_{EMRA} cells and senescent T cells. Effector T cells generated during normal immune responses, terminally differentiated T effector memory CD45RA cells (T_{EMRA} cells), which are mostly responsive to latent viruses, and senescent T cells are all characterized by their excessive production of pro-inflammatory mediators. However, the signalling and transcriptional control mechanisms in the different cell states differ. For effector T cells, ligation of T cell receptor (TCR) and CD28 induces a signalling cascade that activates the mammalian target of rapamycin complex 1 (mTORC1) pathway as well as several transcriptional activators. For T_{EMRA} cells, sestrins provide a scaffold for broad mitogen-activated protein kinase (MAPK) activation, in particular activation of p38 MAPK by the AMP-activated protein kinase (AMPK)–TAB1 complex, while mTORC1 is inhibited. For senescent T cells, DNA damage and cellular stress activate signalling pathways, with nuclear factor- κ B (NF- κ B), p38, CCAAT-enhancer-binding protein- β (C/EBP β) and mTORC1 playing critical roles in regulating the senescence-associated secretory phenotype (SASP). AP-1, activator protein 1; cGAS, cyclic GMP–AMP synthase; ERK, extracellular signal-regulated kinase; IRF, interferon regulatory factor; JNK, JUN N-terminal kinase; NFAT, nuclear factor of activated T cells; PKC, protein kinase C; STING, stimulator of interferon genes protein.

T cells owing to increased expression of survival factors or reduced clearance by innate immune cells. Consistent with the concept that senescent cells exert systemic detrimental effects, T_{EMRA} cells have been implicated in several chronic disease states as well as in poor vaccine responses^{122–124}. However, expanded T_{EMRA} cells have been shown to have both positive and negative effects on life expectancy¹²⁵.

Nevertheless, it should be noted that T_{EMRA} cells exhibit clear differences to true senescent cells. Most importantly, their cell cycle arrest is reversible¹²⁶. In support of the notion that they are a form of terminal differentiation, they gain not only secretory features but also the expression of regulatory cell surface receptors such as leukocyte immunoglobulin-like receptors and killer cell immunoglobulin-like receptors^{127–129}. Their function depends on several biological pathways, including a reduced expression of NAD-dependent protein deacetylase sirtuin 1, which leads to increased lysosomal degradation of FOXO1, and metabolic reprogramming, which favours glycolytic activity and granzyme B production¹³⁰. A characteristic hallmark of these cells is the activation of p38 mitogen-activated protein kinase (MAPK) downstream of AMP-activated protein kinase (AMPK) and TAB1, independent of TCR signalling^{131,132}, thereby uncoupling T cell activation from antigen recognition (FIG. 5). Inhibition of p38 MAPK restores the telomerase activity and proliferative potential of T_{EMRA} cells. More recent studies have shown that, in old T_{EMRA} cells, JUN N-terminal kinase and extracellular signal-regulated kinase are activated downstream of sestrins and this activation

was independent of regulators of the upstream MAPK cascade^{131,133}.

Targeting senescent T cells. The idea that even infrequent senescent immune cells could have harmful system-wide effects and that clearance of senescent cells may be ineffective in ageing has led to the development of therapeutic approaches to eliminate senescent cells¹³⁴. Several compounds — known as *senolytics* — have been developed to exhibit various degrees of selectivity in inducing death of senescent cells by targeting apoptosis, chaperones and histone modifications^{99,134}. The best-examined senolytics are a combination of dasatinib and quercetin, which has shown benefits in model systems of cardiovascular disease and osteoarthritis as well as increases in lifespan^{135–137}. However, senolytics have not been explored to improve immune function, and it is unclear whether they would deplete T_{EMRA} cells and, if so, whether they would be harmful or beneficial. As T_{EMRA} cells maintain effector functionality and are specific for latent viruses, their depletion may enable viral reactivation. In fact, memory inflation by T_{EMRA} cells may account for the successful control of latent cytomegalovirus infection in the elderly. In line with this, infection with VZV, which is also a herpesvirus, does not induce memory inflation of T_{EMRA} cells and shows frequent reactivation, presenting as shingles¹³⁸.

An alternative approach to interfere with the harmful functions of senescent cells is to inhibit the production or activity of pro-inflammatory mediators¹³⁹ (FIG. 5). Treatment of senescent preadipocytes with a pan-Janus kinase (JAK) inhibitor reduced the

Senolytics
Pharmacological compounds that preferentially deplete senescent cells.

production of pro-inflammatory mediators in vivo, possibly by interfering with a positive feedback loop from secreted cytokines¹⁴⁰. Moreover, treatment of aged mice with a JAK1/JAK2 inhibitor alleviated SASP and frailty. However, T cell homeostasis depends on JAK–signal transducer and activator of transcription 5 (STAT5) signalling, and other JAK–STAT pathways are involved in T cell differentiation and function, indicating the limitation of this intervention. Indeed, the risk of developing shingles due to VZV reactivation was increased in patients with rheumatoid arthritis who were treated with the JAK1/JAK3 inhibitor tofacitinib^{141,142}. A similar increase in incidence of shingles was seen in patients treated with the JAK1/JAK2 inhibitor baricitinib¹⁴³.

SASP inhibitors mostly target the nuclear factor- κ B, mTORC and p38 MAPK pathways, which are active in senescent cells. In vitro studies, low-dose inhibition of both mTORC1 and mTORC2 using the mTOR-specific ATP mimetic AZD8055 reversed major phenotypes of senescence in near-senescent fibroblasts¹⁴⁴. In a recent phase IIa study, a cohort of older individuals was treated for 6 weeks with a combination of a catalytic and an allosteric mTOR inhibitor in low doses to selectively inhibit mTORC1 (REF.¹⁴⁵). Treated individuals had an improved response to influenza virus vaccination, without any major side effects, including no reactivation of latent viruses. It is possible that mTORC1 inhibition only modified the vaccine response without targeting senescent cells. In animal experiments, inhibition of the mTORC pathway has been shown to favour CD8⁺ T cell memory over effector T cell generation¹⁴⁶. Moreover, mTORC inhibitors could counteract the sustained signalling in CD4⁺ T cell responses from older individuals arising due to repression of the negative regulators phosphatase and tensin homologue and Sprouty 1, thereby favouring transcription factor networks involved in T follicular helper cell and memory T cell differentiation⁹⁴. However, treated individuals also had significantly fewer infections over the subsequent year; this extended benefit following a short treatment period is surprising and indicated a sustained effect, possibly by eliminating senescent cells.

Other recent approaches are based on AMPK-dependent MAPK activation being central for senescence in T_{EMRA} cells¹³¹. This implies that inhibition of mTORC1, which is in many aspects antagonistic to AMPK, could be harmful; the more appropriate treatment approaches would be the transcriptional repression of sestrins or inhibition of downstream p38 activation.

Inhibition of p38 MAPK has been shown to partially restore proliferative capacity in T_{EMRA} cells by inducing telomerase activity¹⁴⁷.

Conclusions and future perspectives

Pathways implicated in the ageing process, and in particular in stem cell ageing, are highly relevant for the T cell system. T cells depend on well-regulated replenishment and homeostasis, they need to maintain quiescence while being poised to respond and, as highly proliferative cells, they are at risk of developing cellular senescence. However, studies in healthy elderly individuals have shown that ageing-associated differences are not extreme and are frequently within the range of general population variation; accordingly, many healthy older individuals are immunocompetent. A recent extensive immune profiling study in a population selected for absence of major comorbidities and low frailty indices identified only a few immune markers that correlated with age²⁸. Major immune defects do not appear to be an inevitable consequence of ageing.

Studies into T cell ageing therefore provide an opportunity to understand how challenges can be overcome and healthy T cell ageing be supported. The central theme that emerges is to promote cellular quiescence, for example, by reducing infectious burden or by lessening endogenous and exogenous inflammatory stimuli. Many of the T cell-intrinsic defects compromising T cell activation and differentiation in ageing arise from the activation of normal differentiation pathways rather than from the induction of irreversible cellular senescence. If the overall T cell compartment structure is largely intact, as appears to be the case in many healthy older individuals who have reduced but sufficient T cell numbers and diversity, these molecular defects represent druggable targets to improve the efficacy of vaccinations. However, there is scarce data on whether this also applies to T cells in less healthy older individuals. It is possible that frail individuals have major immune defects, such as a major diminution of cell numbers or contraction of diversity. Alternatively, various small changes that might act together to cause a clinical immune defect. Studies into systems immunology aimed at understanding the synergistic impact of small effects, such as naturally occurring variation in immune parameters, on immune responses are still in their infancy but will eventually provide insight into how to interpret and target ageing-associated changes.

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